

CAN YOU STEP IN THE SAME RIVER TWICE?

An ancient Greek philosopher named Heraclitus (535–475 BCE) first asked himself the question about 2,500 years ago and came up with this response, “You could not step twice into the same river; for other waters are ever flowing onto you.” The water in the river is always rushing forward, so although it looks the same, by the time you place your foot in the river on a second occasion, it is different. Of course, you could answer the question another way. Even though the water is always changing, the river itself remains the same—the Thames is always called the Thames, the Mississippi remains the Mississippi, and the Ganges is still the Ganges. In this sense, you can step in the same river as many times as you like, but it is not the same water wetting your feet each time.

CONSTANT CHANGE

Heraclitus was making a point about change. He believed that everything is always changing, and he was right. For example, the cells in our bodies are replaced countless times during our lifetime. If they did not, our bodies could not grow and function. Yet we remain the same person at all stages of our life, whether a baby, teenager, or senior citizen. Similarly, we live on a restless (dynamic) planet. Even an object as apparently fixed and unchangeable as rock is always on the move. Molten rock (magma) makes its way through cracks

in Earth’s crust, where it cools and solidifies to form granite rocks. Wind and rain erodes the rocks, and they are washed into rivers as gravel and sand before ending up on the seabed. Earth’s crust is divided into huge floating plates. When they collide into one another, layers of rock push up to form mountain ranges. Around 30 million years ago, India crashed into Asia to form the Himalaya Mountains, and it is still moving north at a rate of 2 in. (5 cm) per year. That’s twice as fast as human fingernails, which grow about 0.9 in. (2.3 cm) per year.

WATER RECYCLING

Water (H₂O) can change its state from a liquid to a solid (ice) or a gas (vapor) without altering its molecular structure. Most of the water on Earth probably erupted from volcanoes about four billion years ago, collecting as vapor in the atmosphere. Water is continually being moved around the planet. The Sun’s heat turns water from the oceans into vapor, which rises into the air, where it cools and forms clouds made of tiny water droplets. Winds blow the clouds across the land, and they release their moisture as rain or snow to fill lakes and rivers that drain into the oceans. This brings us back to the river you stepped in earlier. In the words of American philosopher Ralph Waldo Emerson (1803–1882), “We change, whether we like it or not.”